

20

Question 1 (12 Marks)

Marks

(a) Evaluate $\sqrt{\frac{53.9 \times 17.8}{19.6 + 4.97}}$ correct to 2 decimal places.

1

(b) Solve the following equations:

(i) $8^x = \frac{1}{4}$

2

(ii) $\frac{6}{x-1} - \frac{3}{x} = 2$

3

(c) Find a primitive of $8 + \frac{3}{e^x}$

2

(d) Simplify: $\frac{2\sqrt{3}}{\sqrt{3}-1}$

2

(e) A store makes a profit of 40% on the cost of all its sales. Find

(i) the selling price if the cost price is \$84

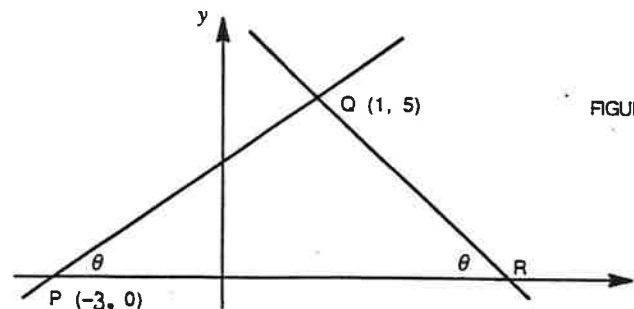
1

(ii) the cost price if the selling price is \$84.

1

Question 2 (12 Marks)

Marks



In the diagram P and Q have coordinates (-3,0) and (1,5) respectively, and $\angle QPR = \angle QRP = \theta$.

Copy this diagram onto your ANSWER SHEET.

(a) Find the coordinates of the midpoint of PQ.

1

(b) Show that the gradient of PQ is $\frac{5}{4}$.

1

(c) Show that the equation of PQ is $5x - 4y + 15 = 0$.

2

(d) Show that the gradient of QR is $-\frac{5}{4}$.

1

(e) Show that the equation of QR is $5x + 4y - 25 = 0$.

2

(f) Find the coordinates of R.

1

(g) Hence find the perpendicular distance from R to PQ.

2

(h) On your diagram, shade in the region satisfying both the inequalities: $5x - 4y + 15 \leq 0$ and $5x + 4y - 25 \geq 0$ simultaneously.

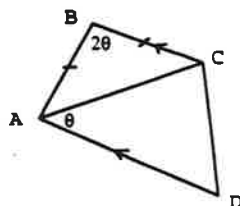
2

Question 3 (12 Marks)

Start a new page.

Marks

(a)

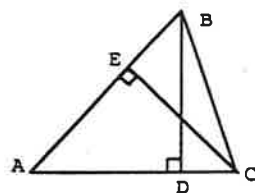


In the diagram, ABCD is a quadrilateral with $AB = BC$, $\angle ABC = 2\theta$ and $\angle CAD = \theta$.

Copy this diagram onto your ANSWER SHEET and find the value of θ , giving reasons.

3

(b)



In the diagram $BD \perp AC$ and $CE \perp AB$.

(i) Copy this diagram onto your ANSWER SHEET and prove that $\triangle ECA$ is similar to $\triangle DBA$.

2

(ii) If $AB = 10\text{cm}$, $BD = 7\text{cm}$ and $AC = 16\text{cm}$, find the length of CE .

2

Question 3 continues over page

Question 3 continued

(c)

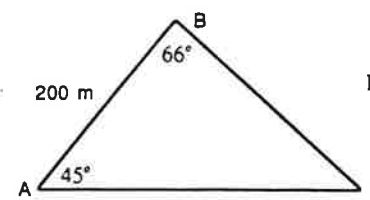


FIGURE NOT TO SCALE

Use the Sine Rule to calculate the length of the side BC to the nearest metre.

2

(d) Solve the equation: $2\sin\theta - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

3

Question 4 (12 Marks) Start a new page.

Marks

(a) Integrate with respect to x :

(i) $\int (2x-1)^6 dx$ 1

(ii) $\int \sin 2x dx$ 1

(b) Integrate and leave answer in exact form: $\int_0^2 e^{2x} dx$ 2

(c) Find the exact area of the region bounded by the curve $y = \frac{1}{x+4}$, the x -axis and the lines $x = 0$ and $x = 4$, giving your answer in the simplest form.

3

(d) (i) Differentiate $y = x^2 e^{-3x}$ with respect to x . 2

(ii) Find the equation of the tangent to the curve $y = x^2 e^{-3x}$ at $x = \frac{1}{3}$.
Leave your answer in exact form.

3

Question 5 (12 Marks)

Start a new page.

Marks

(a) Given that α and β are the roots of the quadratic function $2x^2 - 3x - 5 = 0$ find

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $(\alpha + 1)(\beta + 1)$ 1

(b) Prove that the line $y = 4x - 4$ is a tangent to $y = x^2$. 2

(c) The mass M kg of a radioactive substance present after t years is given by $M = 10e^{-kt}$, where k is a positive constant. After 100 years the mass has reduced to 5 kg.

(i) What was the initial mass? 1

(ii) Find the value of k (leave answer in exact form). 2

(iii) What amount of the radioactive substance would remain after a period of 1000 years? Give answer correct to 4 significant figures.

2

(iv) How long would it take for the initial mass to reduce to 8 kg? Give answer in terms of years, correct to 2 decimal places.

2

Question 6 (12 Marks) Start a new page.

Marks

(a) An office worker is employed at an initial salary of \$20 600 p.a. After each year this salary is increased by \$800.

(i) What is the salary for the seventh year of service? 2

(ii) What is the worker's total salary for the first seven years? 2

(b) Find the number "n" which when added to each of 2, 5 and 9 will give a set of three numbers in geometric progression.

2

(c) Show that the following represents an arithmetic series and hence evaluate

$$\sum_{k=1}^9 (4k - 1)$$

2

(d) A ball is dropped from a height of 20cm and continues to bounce $\frac{3}{4}$ of the height of the preceding bounce until it comes to rest. What is the total distance travelled by the ball?

4

Question 7 (12Marks) Start a new page.

Marks

(a) The minute hand of a clock is 20cm long.

(i) Show that the arc length along which the tip of the hand travels in 16 minutes is $\frac{32}{3}\pi$ radians. 1

(ii) Calculate the shortest distance between the initial and final positions of the tip of the hand. Give answer correct to 2 decimal places.

2

(b)

x	1	2	3	4	5
f(x)	0	0.3	0.5	0.6	0.7

Given the above table, where $f(x) = \log_{10} x$, use Simpson's Rule with the 5 function values to find the approximate area under the curve between $x = 1$ and $x = 5$ (correct to 1 decimal place).

2

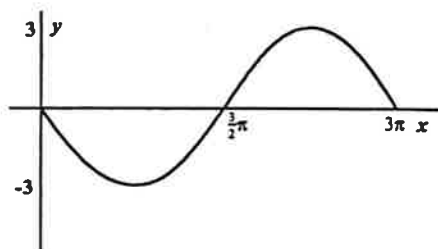
(c) A sheep, grazing in a paddock, is tethered to a stake by a rope 20m long. If the stake is 10m from a fence, find the area over which the sheep can graze. Give answer correct to 2 decimal places.

4

Question 7 continues over page

Question 7 continued

(d)



The above diagram represents a possible sine or cosine curve.

- | | |
|---|---|
| (i) Give the amplitude. | 1 |
| (ii) Give the period. | 1 |
| (iii) Write down a possible equation for the curve. | 1 |

Question 8 (12 Marks) Start a new page.

Marks

- | | | |
|-------|--|---|
| (a) | Given the function $y = x^3 + 3x^2 - 24x + 1$ | |
| (i) | find any maximum or minimum turning points | 4 |
| (ii) | find any points of inflexion | 2 |
| (iii) | determine where the graph cuts the y-axis | 1 |
| (iv) | hence sketch the curve indicating its essential features.
Note: x intercepts not required. | 2 |
| (b) | Jane makes equal annual contributions of \$P into a retirement fund. This retirement fund pays compound interest of 8% per annum compounded annually. The first contribution was made on the 1 st January 2001 and the last contribution is to be made on the 1 st January 2025. | |
| (i) | How much does the first contribution amount to at the end of n years after interest is paid? | 1 |
| (ii) | If Jane wants to retire with a lump sum of \$200 000 on the 31 st December 2025 after interest is paid for the year, find the amount of each equal annual contribution \$P. | 2 |

Question 9 (12 Marks) Start a new page.

Marks

(a) Solve for x : $9^x - 10.3^x + 9 = 0$

2

(b) A particle initially at rest at the origin moves in a straight line with velocity v metres per second, such that $v = 3t(4 - t)$ where t is the time elapsed in seconds. Find:

(i) the acceleration of the particle at the end of 1 second

1

(ii) an expression for the displacement x of the particle in terms of t

2

(iii) the particle's displacement when it is next at rest

2

(iv) the velocity of the particle when it returns to the origin and state the direction in which the particle is travelling then?

2

(v) the time taken for the particle to reach its greatest velocity

1

(vi) the distance travelled by the particle in the first 5 seconds

2

Question 10 (12 Marks)

Start a new page.

Marks

(a)

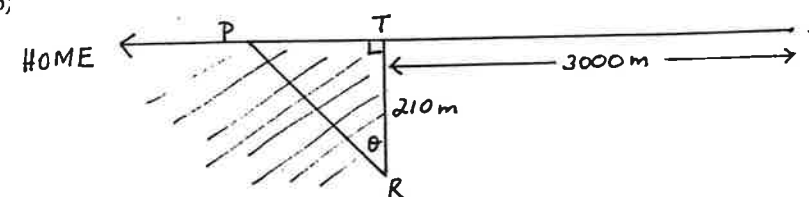
(i) Sketch the curve $y = e^{-2x}$ and shade the region bounded by this curve, the x -axis, the y -axis and the line $x = 2$.

1

(ii) Find the volume generated when this area is rotated about the x -axis (give your answer in terms of π).

3

(b)



The diagram above shows a restaurant R which is 210 m along a road to a point T on the main street. There is a road TR and a grassed area to walk across at an angle θ from the restaurant to this street. Samantha must catch the last bus home after having dinner at this restaurant. The bus travels from S to T to P and beyond to home. The bus will leave the bus stop S at 10 pm sharp, travelling at a speed of 18 m s^{-1} . The distance from S to T is 3000 m. Samantha decides to walk across the grass at angle θ and reaches the street at the point P such that $\angle PRT = \theta$. She walks at a speed of 3 m s^{-1} .

(i) Find PR and PS in terms of θ .

2

(ii) Find two expressions in terms of θ , one expression for the time taken, in seconds, for the bus to travel from S to P and the other expression for the time taken, in seconds, by Samantha to walk from R to P .

2

(iii) What is the latest that Samantha can leave the restaurant in order to catch this bus? Do not test for maximum or minimum, as we are told that there is a maximum only.

4

End of paper

2 UNIT TRIAL Solutions

Question 1 (a) 6.25 (to 2dp) 1

(b) (i) $8^x = \frac{1}{4}$
 $2^{3x} = 2^{-2}$ 1
 $3x = -2$ 1
 $x = -\frac{2}{3}$

(ii) $\frac{6}{x-1} - \frac{3}{x} = 2$

$6x - 3(x-1) = 2x(x-1)$ 1

$6x - 3x + 3 = 2x^2 - 2x$ 1

$2x^2 - 5x - 3 = 0$ 1

$(2x+1)(x-3) = 0$

$x = -\frac{1}{2} \text{ or } 3$ 1

(c) $\frac{2\sqrt{3}}{\sqrt{3}-1} = \frac{2\sqrt{3}}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$ $\frac{1}{2}$

$= \frac{2\sqrt{3}(\sqrt{3}+1)}{3-1}$ $\frac{1}{2}$

$= \frac{2\sqrt{3}(\sqrt{3}+1)}{2}$ $\frac{1}{2}$

$= \sqrt{3}(\sqrt{3}+1)$ $\frac{1}{2}$
 $= 3 + \sqrt{3}$ $\frac{1}{2}$

(i) S.P = $1.4 \times \$84$
 $= \$117.60$

(ii) $140\% = \$84$
 $1\% = \frac{84}{140}$
 $100\% = \frac{84}{140} \times 100$
 $= \$60$

Q.2 (contd)

(d) $m(QR) = ?$

$$\tan \theta = \frac{5}{4}$$

$$\therefore \tan(180^\circ - \theta) = -\tan \theta = -\frac{5}{4}$$

(e) Epr. of QR: $y - y_1 = m(x - x_1)$
 $y - 5 = -\frac{5}{4}(x - 1)$
 $4y - 20 = -5x + 5$
 $5x + 4y - 25 = 0$

(f) R lies on QR $\rightarrow 5x + 4y - 25 = 0$
 when $y = 0$, $x = ?$

$$5x - 25 = 0$$

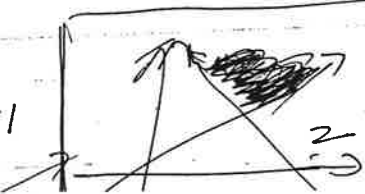
$$5x = 25$$

$$x = 5$$

$$\therefore R = (5, 0)$$

(g) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|5 \times 5 + (-4) \times 0 + 15|}{\sqrt{25 + 16}}$
 $= \frac{|40|}{\sqrt{41}} = \frac{40}{\sqrt{41}} \text{ units}$

$$\left. \begin{array}{l} a=5, b=-4, c=15 \\ x_1=5, y_1=0 \end{array} \right\}$$



Question 3

(a) $\angle BAC = \angle ABC = \frac{180^\circ - 2\theta}{2}$ (base $\angle$'s of $\triangle ABC$ are equal and angles of $\triangle = 180^\circ$)

$$\angle ABC + \angle BAD = 180^\circ \text{ (Co. int } \angle \text{'s, } BC \parallel AD \text{)}$$

$$\therefore 2\theta + \frac{180^\circ - 2\theta}{2} + \theta = 180^\circ$$

$$4\theta + 180^\circ - 2\theta + 2\theta = 360^\circ$$

$$\therefore 4\theta = 180^\circ$$

$$\theta = 45^\circ$$

(b) (i) In $\triangle ECA, DBA$

(i) $\angle AEC = \angle ADB$ (both 90° , given)

(ii) $\angle EAC = \angle BAD$ (common)

$\therefore \triangle ECA \parallel \triangle DBA$ (equiangular)

(ii) $\frac{CE}{BD} = \frac{AC}{AB}$ (Corr. sides of similar $\triangle$'s are in the same ratio)

$$\therefore \frac{CE}{7} = \frac{16}{10}$$

$$\therefore CE = \frac{16 \times 7}{10} = \frac{112}{10} = 11.2$$

Qu. 3 (contd)

$$(c) \quad \frac{BC}{\sin 45^\circ} = \frac{200}{\sin 69^\circ}$$

$$(\angle BCA = 180^\circ - (45^\circ + 66^\circ))$$

angle sum of $\triangle$

$$\therefore BC = \frac{200 \times \sin 45^\circ}{\sin 69^\circ}$$

$$= 151.148 \dots$$

$$= 151 \text{ m (to nearest m)}$$

$$(d) \quad 2 \sin \theta - 1 = 0$$

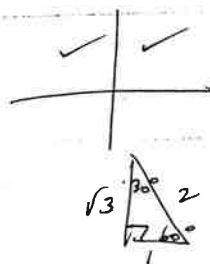
$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\text{acute } \theta = 30^\circ$$

$$\therefore \theta = 30^\circ, 150^\circ$$

$$0^\circ \leq \theta \leq 360^\circ$$



Question 4

$$(a) (i) \quad \int (2x-1)^6 dx$$

$$= \frac{(2x-1)^7}{7 \times 2} + C$$

$$= \frac{(2x-1)^7}{14} + C$$

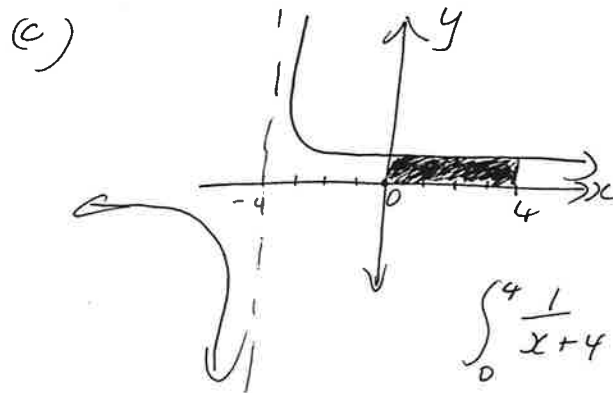
$$(ii) \quad \int \sin 2x dx$$

$$= -\frac{\cos 2x}{2} + C$$

$$(b) \quad \int_0^2 e^{2x} dx = \left[\frac{e^{2x}}{2} \right]_0^2$$

$$= \frac{e^4}{2} - \frac{e^0}{2}$$

$$= \frac{e^4}{2} - \frac{1}{2}$$



$$\int_0^4 \frac{1}{x+4} dx = \left[\ln(x+4) \right]_0^4$$

$$= \ln 8 - \ln 4$$

$$= 3 \ln 2 - 2 \ln 2$$

$$= \ln 2$$

Qu. 4 (contd.)

(d) (i)

$$y = x^2 e^{-3x}$$

$$y' = x^2 \cdot -3e^{-3x} + 2x \cdot e^{-3x}$$

$$y' = xe^{-3x} (2 - 3x)$$

(ii)

$$y = x^2 e^{-3x}, \quad y' = xe^{-3x} (2 - 3x)$$

$$\text{When } x = \frac{1}{3}, \quad y = \frac{1}{9} e^{-1} = \frac{1}{9e} \quad \frac{1}{2}$$

$$\text{and } y' = m = \frac{1}{3} e^{-1} (2 - 1) = \frac{1}{3e} \quad \frac{1}{2}$$

$\therefore$ Eqn of tangent to curve at $x = \frac{1}{3}$ is:

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{9e} = \frac{1}{3e} (x - \frac{1}{3}) \quad \frac{1}{2}$$

$$9ey - 1 = 3x - 1$$

$$y = \frac{3x}{9e}$$

$$\therefore y = \frac{x}{3e} \quad \frac{1}{2}$$

Qu. 5 (contd.)

(c)

$$M = 10e^{-kt}$$

(i) When $t = 0$, $M = ?$

$$M = 10e^0$$

$$M = 10$$

$\therefore$ initial mass is 10 kg.

(ii)

When $t = 100$, $M = 5$

$$\therefore 5 = 10e^{-100k}$$

$$e^{-100k} = 0.5 \quad \frac{1}{2}$$

$$\ln e^{-100k} = \ln 0.5 \quad \frac{1}{2}$$

$$\therefore -100k = \ln 0.5 \quad \frac{1}{2}$$

$$k = \frac{\ln 0.5}{-100} \quad \frac{1}{2}$$

(iii)

When $t = 1000$, $M = ?$

$$M = 10e^{-1000k}$$

$$= 10e^{-1000 \times \frac{\ln 0.5}{-100}}$$

$$= 0.009765625 \text{ (kg)} \quad \frac{1}{2}$$

$$= 9.766 \text{ g (4 s.f.)} \quad \frac{1}{2}$$

(iv)

When $M = 8$, $t = ?$

$$8 = 10e^{-kt}$$

$$e^{-kt} = 0.8$$

$$\ln e^{-kt} = \ln 0.8$$

$$-kt = \log 0.8$$

$$t = \frac{\log 0.8}{-k}$$

$$= 32.1940$$

Question 5

(a) $2x^2 - 3x - 5 = 0$

(i) $\alpha + \beta = -\frac{b}{a} = \frac{3}{2}$

(ii) $2\beta = \frac{c}{a} = -\frac{5}{2}$

(iii) $(\alpha+1)(\beta+1)$

$= \alpha\beta + \alpha + \beta + 1$

$= \alpha\beta + (\alpha + \beta) + 1$

$= -\frac{5}{2} + \frac{3}{2} + 1$

$= 0$

(b)

$y = 4x - 4$, $y = x^2$

Solve simultaneously

$4x - 4 = x^2$

$x^2 - 4x + 4 = 0$

$(x-2)(x-2) = 0$

$\therefore x = 2$

only one root

$\therefore y = 4x - 4$ touches $y = x^2$

$\therefore y = 4x - 4$ is a tangent to $y = x^2$

Q4 b

(a) This is an A.P. with $a = 20600$
 $d = 800$
 $t = ?$

(i) $T_n = a + (n-1)d$

$T_7 = 20600 + 6 \times 800$
 $= 25400$

(ii) $S_n = \frac{n}{2}(a + l)$

$S_7 = \frac{7}{2}(20600 + 25400)$
 $= \$161000$

(b)

G.P. $\rightarrow 2+n, 5+n, 9+n$

$\therefore \frac{n+5}{n+2} = \frac{n+9}{n+5}$

$(n+5)^2 = (n+2)(n+9)$

$n^2 + 10n + 25 = n^2 + 11n + 18$
 $7 = n$

$\therefore n = 7$

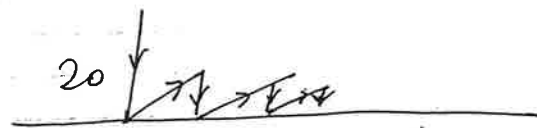
(c)

$\sum_{k=1}^9 (4k-1) = 3 + 7 + 11 + 15 + \dots$

This an AP with $a = 3, d = 4, n = 9$

$\therefore S_9 = \frac{9}{2}(6 + 8 \times 4)$
 $= \frac{9}{2}(38) = 171$

(d) This a G.P. with $a = 15$, $r = \frac{3}{4}$ |



$$\frac{3}{4} \times 20 = 15$$

$$\therefore a = 15, r = \frac{3}{4}$$

$$S_{\infty} = \frac{15}{\frac{1}{4}} = 60$$

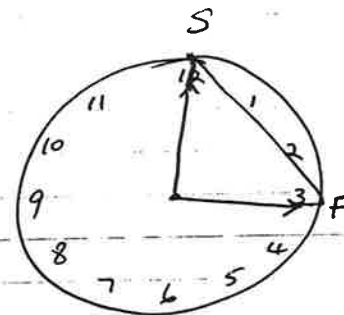
$$\therefore \text{Total distance} = 2 \times S_{\infty} + 20$$

$$= 2 \times 60 + 20$$

$$= 140 \text{ cm.}$$

Question 7

(a)(i)



$$60 \text{ min} = 360^\circ$$

$$1 \text{ min} = \frac{360}{60} = 6^\circ$$

$$\therefore 16 \text{ min} = 16 \times 6 = 96^\circ \quad \frac{1}{2}$$

$$\begin{aligned} \text{Arc length} &= \frac{96}{360} \times 2 \times \pi \times 20 \\ &= \frac{32\pi}{3} \text{ cm.} \end{aligned} \quad \frac{1}{2}$$

(ii) Shortest distance is SF.

$$\begin{aligned} SF^2 &= 20^2 + 20^2 - 2 \times 20 \times 20 \times \cos 96^\circ \\ &= 883.62277 \dots \end{aligned}$$

$$\therefore SF = 29.73 \text{ cm.}$$

7(b) $h = \frac{5-1}{4} = \frac{4}{4} = 1$ or $h = 1$ from $\frac{1}{2}$ table.

$$\therefore A = \frac{1}{3} (0 + 4 \times 0.3 + 2 \times 0.5 + 4 \times 0.6 + 0.7)$$

$$\begin{aligned} A &= \frac{5.3}{3} = 1.777\dots \\ &= 1.8 \text{ unit}^2 \end{aligned} \quad \frac{1}{2}$$

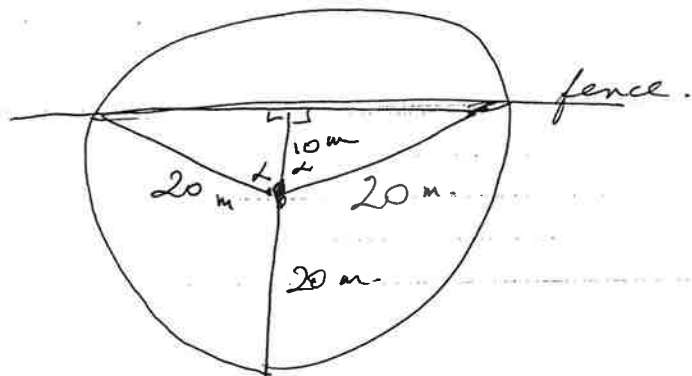
7(c) next sheet

7(d) (i) amplitude = 3

(ii) period = 3π

(iii) $-3 \sin 2x$

(c)



$$\cos \alpha = \frac{10}{20} = \frac{1}{2}$$

$$\alpha = 60^\circ$$

$$2\alpha = 120^\circ = \frac{2\pi}{3}$$

 $\frac{1}{2}$ Let $\theta = 2\alpha$.

$$A \text{ of sector} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 20^2 \times \frac{2\pi}{3} = \frac{400\pi}{3}$$

$$A \text{ of triangle} = \frac{1}{2} ab \sin \theta$$

$$= \frac{1}{2} \times 20 \times 20 \times \sin 120^\circ$$

$$= \frac{1}{2} \times 20 \times 20 \times \frac{\sqrt{3}}{2}$$

$$= 100\sqrt{3}$$

$$A \text{ of segment} = \frac{400\pi}{3} - 100\sqrt{3}$$

$$\text{Area of circle} = \pi r^2 = \pi \times 400 = 400\pi$$

$$\begin{aligned} \therefore \text{grazing area} &= 400\pi - \left(\frac{400\pi}{3} - 100\sqrt{3} \right) \\ &= 1200\pi - 400\pi - 300\sqrt{3} \\ &= 800\pi - 300\sqrt{3} \end{aligned}$$

Question 8

$$(a) (i) \quad y = x^3 + 3x^2 - 24x + 1$$

$$y' = 3x^2 + 6x - 24$$

Stat pts when $y' = 0$

$$\text{i.e. } 3x^2 + 6x - 24 = 0$$

$$(\div 3) \quad x^2 + 2x - 8 = 0$$

$$(x-2)(x+4) = 0$$

$$\therefore x = 2 \text{ or } -4$$

$$y'' = 6x + 6$$

When $x = 2$, $y'' = 12 + 6$ (+ve) $\therefore$ a min. tp.When $x = -4$, $y'' = -24 + 6$ (-ve) $\therefore$ a max. tp.

$$\text{When } x = 2, y = -27$$

$$\text{When } x = -4, y = 81$$

 $\therefore$ a max. tp at $(-4, 81)$ and
a min. tp at $(2, -27)$.
(ii) Pt. of inflexion when $y'' = 0$ and there is a change of concavity.

$$\text{i.e. } 6x + 6 = 0$$

$$6x = -6$$

$$x = -1$$

$$\text{When } x = -1, y = 27$$

x	-1^+	-1	-1^-
y''	$-$	0	$+$

 $\therefore$ a change in concavity $\therefore$ a pt. of inflexion at $(-1, 27)$

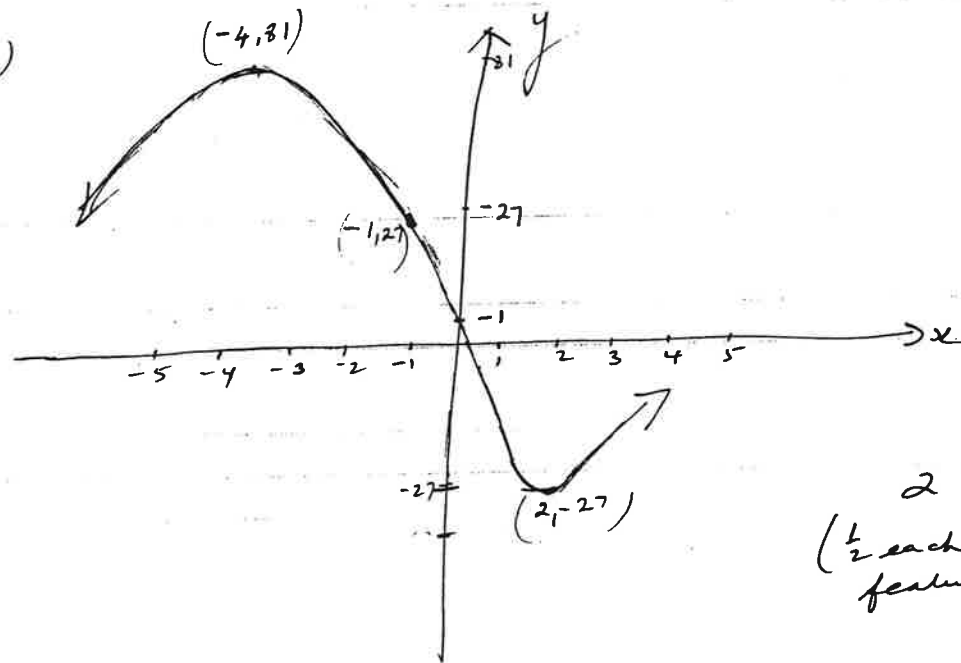
Qu. 8 contd.

(iii) When $x = 0$, $y = ?$

$$y = 1$$

$\therefore$ graph cuts y -axis at $(0, 1)$.

(iv)



2
($\frac{1}{2}$ each
feature)

Qu. 8 contd

8 (b) 1^{25}

$$= \$m (1.08)^{25}$$

2^{nd}

$$= \$m (1.08)^{24}$$

25^{th}

$$= \$m (1.08)^1$$

$$200000 = m [1.08 + 1.08^2 + \dots + 1.08^{25}]$$

$$200000 = m (1.08) \frac{[1.08^{25} - 1]}{1.08 - 1}$$

$$\therefore m = \frac{200000 \times 0.8}{(1.08)(1.08^{25} - 1)}$$

$$m = \$2533.11$$

Question 9

(a) $9x^2 - 10 \cdot 3x + 9 = 0$

Let $k = 3x$

$k^2 - 10k + 9 = 0$

$(k-1)(k-9) = 0$

$\therefore k = 1 \text{ or } 9$

$\therefore 3x = 1 \quad \text{or} \quad 3x = 9$

$3x = 3^0$

$\therefore x = 0$

$3x = 3^2$

$x = 2$

$\therefore x = 0 \text{ or } 2$

(b) (i) $v = 3t(4-t)$

$v = 12t - 3t^2$

$\therefore a = 12 - 6t$

When $t = 1$, $a = 12 - 6 = 6 \text{ m/s}^2$

(ii) $x = \int (12t - 3t^2) dt$

$= \frac{12t^2}{2} - \frac{3t^3}{3} + C$

$= 6t^2 - t^3 + C$

When $t = 0$, $x = 0$, $\therefore C = 0$

$\therefore x = 6t^2 - t^3$

(iii) When $v = 0$

$3t(4-t) = 0$

$\therefore t = 0 \text{ or } 4$

$\therefore$ it is next at rest after 4 sec.

$\therefore$ when $t = 4$, $x = 6 \cdot 4^2 - 4^3$

$= 32 \text{ m}$

(iv) When $x = 0$, $6t^2 - t^3 = 0$

$t^2(6-t) = 0$

$t = 0 \text{ or } 6$

$\therefore$ when $t = 6$, $v = 18(4-6)$
 $= -36 \text{ m/s}$

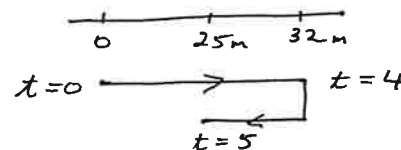
(v) Max velocity when $a = 0$

$\therefore 12 - 6t = 0$

$t = 2$

$\therefore$ it takes 2 sec for it to reach max. velocity.

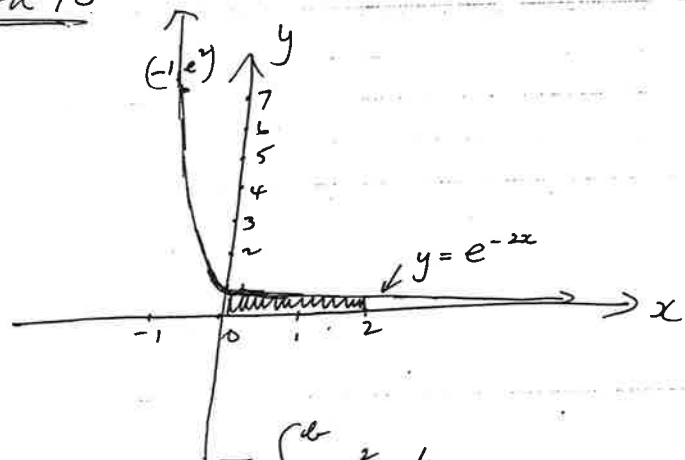
(vi) When $t = 5$, $x = 150 - 125 = 25 \text{ m}$



$\therefore$ it travels $32 + (32 - 25) = 39 \text{ m}$ in the first 5 sec.

Question 10

(a) (i)



2

(ii)

$$V = \pi \int_a^b y^2 dx$$

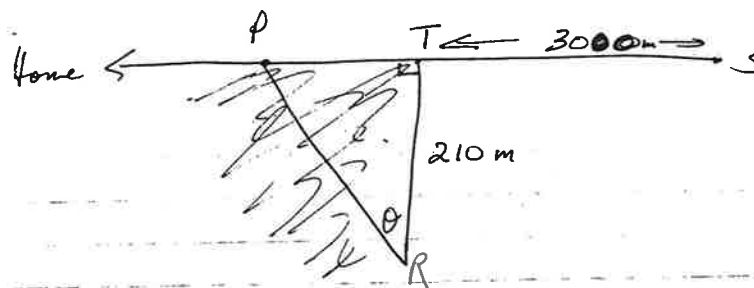
$$V = \pi \int_0^2 (e^{-2x})^2 dx$$

$$= \pi \int_0^2 e^{-4x} dx$$

$$= \pi \left[\frac{e^{-4x}}{-4} \right]_0^2$$

$$= \pi \left(\frac{e^{-8}}{-4} + \frac{e^0}{4} \right)$$

$$= \frac{\pi}{4} \left(1 - \frac{1}{e^8} \right) \text{ units}^3$$



$$S(\text{Samantha}) = 3 \text{ m/s}$$

$$S(\text{bus}) = 18 \text{ m/s}$$

(i)

$$\cos \theta = \frac{210}{PR}$$

$$\therefore PR = \frac{210}{\cos \theta} \text{ or } 210 \sec \theta$$

$$\tan \theta = \frac{PT}{210}$$

$$\therefore PT = 210 \tan \theta$$

$$\therefore PS = 210 \tan \theta + 3000$$

(ii)

$$t(\text{bus}) = \frac{d}{S} = \frac{210 \tan \theta + 3000}{18}$$

$$t(\text{Samantha}) = \frac{d}{S} = \frac{210}{\frac{3 \cos \theta}{1}} = \frac{210 \sec \theta}{3}$$

$$(iii) T = t(\text{bus}) - t(\text{Samantha})$$

$$= \frac{210 \tan \theta + 3000}{18} - \frac{210 \sec \theta}{3}$$

$$= \frac{210 \tan \theta + 3000 - 1260 \sec \theta}{18}$$

$$\begin{aligned}\frac{dT}{d\theta} &= \frac{1}{18} (210 \sec^2 \theta - 1260 \sec \theta \tan \theta) \\ &= \frac{35 \sec^2 \theta - 210 \sec \theta \tan \theta}{3} \\ &= 35 \sec \theta \left(\frac{\sec \theta}{3} - 2 \tan \theta \right)\end{aligned}$$

When $\frac{dT}{d\theta} = 0$,

$$35 \sec \theta = 0$$

$$\text{im } \sec \theta = 0 \quad (\text{no soln})$$

or $\frac{\sec \theta}{3} - 2 \tan \theta = 0$

$$\frac{\sec \theta}{3} = 2 \tan \theta$$

$$\frac{1}{3 \cos \theta} = \frac{2 \sin \theta}{\cos \theta}$$

$$\therefore \sin \theta = \frac{1}{6}$$

$$\therefore \theta = 9^\circ 36' \quad (\text{to nearest min})$$

$$\text{or } \theta = 0.1674480 \text{ radians}$$

$$\therefore T = \frac{210 \tan \theta + 3000 - 1260 \sec \theta}{18}$$

$$= \frac{210 \tan 0.16745 + 3000 - 1260 \sec 0.1674}{18}$$

$$= 97.64514 \text{ secs. after } 10 \text{ pm.}$$

$$= 1 \text{ min } 38 \text{ secs. after } 10 \text{ pm} \quad (\text{using radians})$$

or $1 \text{ min } 39 \text{ secs after } 10 \text{ pm} \quad (\text{using degrees})$